Competitive Exam Practice 5: Process Concept and the PCB

GATE (Computer Science) Pattern

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Each question carries 1–2 marks. MCQ = single correct option; NAT = Numerical Answer Type. *All questions in this set are original, written in the GATE pattern; none is claimed to be a past-year question.*

Q1 *[Original, GATE-pattern]* **[MCQ, 1 mark]** Which statement about programs and processes is **TRUE**?

(A) A program and a process are the same entity viewed from disk and from memory

(B) Two processes executing the same program share a single PCB

(C) Two processes executing the same program may share the text section but have separate stacks

(D) A process exists for as long as its executable file exists on disk

Q2 *[Original, GATE-pattern]* **[MCQ, 1 mark]** In the five-state process model, which transition is **NOT** possible?

(A) running $\rightarrow$ ready

(B) waiting $\rightarrow$ running

(C) waiting $\rightarrow$ ready

(D) new $\rightarrow$ ready

Q3 *[Original, GATE-pattern]* **[MCQ, 1 mark]** A local variable declared inside a C function, and the memory returned by `malloc` inside that function, reside respectively in:

(A) data, heap

(B) stack, heap

(C) heap, stack

(D) stack, data

Q4 *[Original, GATE-pattern]* **[MCQ, 2 marks]** Which of the following is **NOT** normally stored in a process control block?

(A) The program counter

(B) Base and limit registers or a page-table pointer

(C) The source code of the program

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(D) The list of open files

Q5 *[Original, GATE-pattern]* **[MCQ, 2 marks]** A running process issues a blocking `read` on the keyboard. Later, the key is pressed and the I/O completes while another process is running. The sequence of states the first process passes through, starting from running, is:

(A) running → waiting → running

(B) running → ready → waiting → running

(C) running → waiting → ready → running

(D) running → terminated → ready → running

Q6 *[Original, GATE-pattern]* **[NAT, 1 mark]** A system with 4 CPUs has 15 processes, none new or terminated. Exactly 6 are in the waiting state. The minimum possible number of processes in the ready state is _____.

Q7 *[Original, GATE-pattern]* **[NAT, 2 marks]** A single-CPU process is created, admitted, dispatched, preempted by the timer, dispatched, blocks for disk I/O, has its I/O complete, is dispatched, is preempted by the timer, is dispatched, and exits. The total number of state transitions it undergoes is _____.

Q8 *[Original, GATE-pattern]* **[NAT, 2 marks]** A PCB stores: a program counter (8 B), 32 general-purpose registers of 8 B each, a status register (8 B), a stack pointer (8 B), and 200 B of other control information. On each interrupt, only the processor-state information (PC, general registers, status register, stack pointer) is saved. The number of bytes saved per interrupt is _____.

Q9 *[Original, GATE-pattern]* **[NAT, 2 marks]** One CPU. At some instant: `P1` running; `P2`, `P3` ready; `P4`, `P5`, `P6` waiting. Then, in order: `P1`'s slice expires and `P2` is dispatched; `P2` blocks on I/O and `P3` is dispatched; `P4` and `P5` have their I/O complete. The number of processes in the ready state now is _____.

Q10 *[Original, GATE-pattern]* **[MCQ, 2 marks]** Which reason best explains why the PCB is kept in kernel memory rather than in the process's own address space?

(A) Kernel memory is faster than user memory

(B) The process could otherwise modify its own state, PID, or memory limits, and the kernel must access the PCB even when the process's memory is swapped out

(C) User memory cannot hold structures larger than one page

(D) The PCB is needed only in kernel mode, so its location does not matter

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